MULTI-LEVEL BILL OF MATERIALS
(Bring Roar to Life)

PRODUCT NAME	Knightsaurus (Quadrupedal Robotic Dinosaur)	TEAM MEMBERS
TEAM	Knightsaurus	Dylan Bell
ADVISOR	Dr. Vazquez	Brady Bierwagen
PART COUNT	95	Walker Mitchell
TRUE COST SPENT (EXLCUDING ACQUIRED/OWNED PARTS AND ACCOUNTING FOR 2 PACKS)	$574.59	Liam Russell
TOTAL COST OF PARTS USED	$668.50	Jackson Seery

ITEM NUMBER	ITEM NAME	DESCRIPTION	QUANTITY	UNITS	IMAGE	CATEGORY	ACQUIRED FROM UCF INVENTORY	STATUS	SOURCE	MANUFACTURER	LINK TO SUPPORTING DOC	COMMENTS	UNIT COST	TOTAL PART COST
DS3245SG	45 kg Servo Motor	Waterproof, 8.4v, 270 degrees, Coreles, Steel Gears, Dual Ball Bearing, 51 kg.cm, Stall Current (8.4v): 5.8A	12	1		Quadrupedal Movement	No	Purchased	Amazon	KEAcvise	https://www.amazon.com/KEAcvise-Coreless-Stainless-Waterproof-Robotic-	Sold as a 2 Pack. 12 Needed for the Project (Legs)	$24.00	$287.94
ABX00087	Arduino UNO R4 WIFI Microcontroller	Wifi, Bluetooth, USB-C, ESP32-S3, Renesas RA4M1, Arduino Cloud	1	1		Electronics	No	Owned	Amazon	Arduino	www.amazon.com/Arduino-UNO-WiFi-ABX00087-Bluetooth/dp/B0C8V88Z9D/ref=sr_	Used to control the robot ('brain' of the dinosaur).	$27.50	$27.50
	4S LiPo Battery	14.8v, 6500 mAh, 90 C	1	1		Electronics	Yes	Purchased	Amazon	Nihewo	https://www.amazon.com/Battery-6500mAh-Compatible-Traxxass-Vehicles/dp/B09P82QYNF?th=1	Sold as a 2 pack. Only using 1 battery.	$38.98	$38.98
DS3218MG	20 kg Servo Motor	Waterproof, 8.4v, 180 degrees, 20 kg.cm	2	1		Neck and Tail Movement	No	Purchased	Amazon	AITRIP	https://www.amazon.com/AITRIP-Digital-Torque-Waterproof-Control/dp/B0D1C2THR3?th=1	Used for the neck and tail of the robot. Sold as 2 pack.	$11.00	$21.99
PCA9685	16-Channel Servo Driver	12C input, Voltage DC 5-10 V, 12 bit, 40-1000HZ	1	1		Electronics	No	Purchased	Amazon	Teyleten Robot	https://www.amazon.com/Teyleten-Robot-PCA9685-Compatible-Raspberry/dp/B0CNVBWX2M/ref=	Sold as a 2 pack. Using 3 for the project. Already owned 1.	$19.49	$19.49
LM2596	Buck Converter	4.0v-40v input to 1.25v-37v output, LED Display	3	1		Electronics	No	Purchased	Amazon	HiLetgo	https://www.amazon.com/HiLetgo-Step-down-Converter-1-25-37V-Voltmeter/dp/B00LSEBYHU/ref=sr_	Used to ensure proper voltage for servos.	$5.25	$15.75
	3D Printer Filament (PLA+)	1.75 mm Black 4kg Filament	4	1		Chassis and Legs	No	Purchased	Amazon	ELEGOO	https://www.amazon.com/ELEGOO-Filament-Stronger-Dimensional-Accuracy/dp/B0C7GDB9GQ/ref=		$12.99	$51.96
	Screws	2200PCS M2 M3 M4 M5 Metric Screws Assortment Kit, Hex Socket Button Head Assorted kit, Machine Screw Set with Nuts &	1	1		Hardware	No	Purchased	Amazon	Fgruh	https://www.amazon.com/dp/B0FGX9JGC5/ref=sspa_dk_detail_3?pd_rd_i=B0FGX9JGC5&pd_rd_w=		$19.99	$19.99
6806-2RS	Bearings (30x42x7)	uxcell 6806-2RS Deep Groove Ball Bearings 30mm Inner Dia 42mm OD 7mm Bore Double Sealed Chrome Steel Z2 2pcs	6	1		Hardware	No	Purchased	Amazon	Uxcell	https://www.amazon.com/uxcell-6806-2RS-Groove-Bearings-Double/dp/B082PYMKLG/ref=sr_1	2 Pack of Bearings	$6.39	$38.34
608-2rs	Bearings (8x22x7)	hamiey 50 pcs 608-2rs Ball Bearings 8x22x7mm, Double Rubber Sealed Deep Groove Bearing for 3D Printer, Skateboards, Electric Motor,	1	1		Hardware	No	Purchased	Amazon	Harmiey	https://www.amazon.com/Bearings-8x22x7mm-Skateboards-Electric-	50 Pack of Bearings	$16.99	$16.99
	Speakers	ALAMSCN 3PCS 3 Watt 8 Ohm Speaker Mini Speaker Micro 8ohm 3w Loundspeaker with JST-PH2.0 Interface for Arduino Small Electronic	1	1		Electronics	No	Purchased	Amazon	Ransanx	https://www.amazon.com/ALAMSCN-Loundspeaker-JST-PH2-0-Interface-	Used for enabling sound effects for the dinosaur. Only one used for the project. Comes in a three pack.	$2.33	$2.33
	Dupont Wires (Pin)	120 pcs (40 Male-Male, 40 Female-Male, 40 Female-Female)	1	1		Electronics	No	Purchased	Amazon	ELEGOO	https://www.amazon.com/Elegoo-EL-CP-004-Multicolored-Breadboard-	Used for connecting the electronic components and prototyping.	$6.98	$6.98
	Aluminum Round Tube	27mm OD 25mm Inner Dia 300mm Length	2	1		Hardware	No	Purchased	Amazon	Uxcell	https://www.amazon.com/uxcell-Aluminum-Length-Seamless-Straight/dp/B083HBS4DV/ref=sr_1_		$13.59	$27.18
	Aluminum Square Tube	10mmx10mmx1mm Wall Thickness 300mm	1	1		Hardware	No	Purchased	Amazon	Uxcell	https://www.amazon.com/uxcell-Aluminum-10mmx10mmx1mm-Thickness-		$9.19	$9.19
	Aluminum Round Rod	5/16"(8mm) Diameter	1	1		Hardware	No	Purchased	Amazon	Feelers	https://www.amazon.com/Feelers-Aluminum-Round-Diameter-Length/dp/B0BNQ4Q87H/ref=sr_1		$6.63	$6.63
	Rare Earth Magnet Neodymium Magnet	20x3mm (10 Pack)	4	1		Hardware	No	Purchased	Amazon	Best Choice Magnets	https://www.amazon.com/Magnets-Refrigerator-Neodymium-Whiteboard-	Comes in a 10 pack. Only four used for the project.	$3.60	$14.38
	Square Aluminum Flat Bar	5/16" (8mm) Thick x 5/16" (8mm) Width x 12"; (2 Pack)	1	1		Hardware	No	Purchased	Amazon	Meccanixity	Amazon.com: MECCANIXITY 2 Pcs Square Aluminum Flat Bar, 5/16" (8mm) Thick x 5/16" (8mm) Width x		$7.29	$7.29
	Servo Disk	(20 Pack), Aluminum, 25T	15	1		Hardware	No	Purchased	Amazon	DaFuRui	Amazon.com: DaFuRui 20Pcs Servo Horn Metal Aluminum 25T Silvery Servo Disc for Robot	Sold as a 20 pack. 15 were used for the project.	$0.85	$12.74
	ESP32	Development Board 2.4GHz Dual-Mode WiFi + Bluetooth Dual Cores Microcontroller Processor Integrated with Antenna RF AMP	1	1		Electronics	Yes	Purchased	Amazon	HiLetgo	https://www.amazon.com/HiLetgo-ESP-WROOM-32-Development-Microcontroller-	Used for RC functionalities (*Bluepad32)*	$9.99	$9.99
	Cooling Fan	2Pcs 40mm Fan 5V,3D Printer Micro 5 Volt Fans 4010, Mini Brushless Cooling Fan 40mm x 10mm 2PIN, 5000 RPM,for DIY Cooling	1	1		Electronics	No	Purchased	Amazon	SIXQJZML	https://www.amazon.com/dp/B0DY6GQMJX?ref=cm_sw_r_cso_em_mwn_do_X7XE9MGIDJEE52S2	Only using one for the project. Sold as a 2 pack.	$3.00	$3.00
	Terminal Block	2 Pack 12 Circuits 20-30A 200v-450v Dual Row Screw Terminals Strip +48 PCS Heat Shrink Wire Connectors	1	1		Electronics	No	Purchased	Amazon	GUBCUB	https://www.amazon.com/dp/B09VSVC9DX?ref=cm_sw_r_cso_sms_mwn_dp_QH0QY3R4JWB7XQS0X	Only using one for the project. Sold as a 2 pack.	$5.00	$5.00
GME12864-13	OLED Display	0.96 Inch OLED Module 12864 128x64 Yellow Blue SSD1306 Driver I2C Serial Self-Luminous Display Board for Arduino	1	1		Electronics	No	Purchased	Amazon	UCTRONICS	https://www.amazon.com/UCTRONICS-SSD1306-Self-Luminous-Display-Raspberry/dp/B072Q2X2LL	Used to display battery life and other information. Bought in bulk years ago so the true price is cheaper.	$6.99	$6.99
	LiPo Switch	Current Rating: 35 Amps, Operating Voltage: 240 Volts	1	1		Electronics	No	Purchased	Amazon	Nuofany	https://www.amazon.com/Current-Brushed-Battery-Connector-Fishing/dp/B0BFHZ9XYX	Used to power on and off the electronics.	$9.99	$9.99
	TPU Filament	TPU 3D Printer Filament; 1 Kg	1	1		Feet	No	Purchased	Sunlu	Sunlu	https://store.sunlu.com/products/moq-3kg-tpu-3d-printer-filament-1kg?variant=48044620513562&co	Already owned. Used less than one-fifth of the spool. Price of the entire spool is $14.99.	$3.00	$3.00
	Red LED Diode	5 mm Red LED Diodes, 200 pack (already owned), only used 2.	2	1		Electronics	No	Purchased	Amazon	ZZHXSM	https://www.amazon.com/ZZHXSM-Assorted-Electronic-Components-	Already owned. Used only two LEDs (for the eyes). 200 pack costs $8.49.	$0.08	$0.16
	16-Gauge Wire	16-Gauge Silicon Wire (in Red and Black)	1	1		Electronics	No	Purchased	Amazon	BNTECHGO	https://www.amazon.com/BNTECHGO-Silicone-Flexible-Strands-Stranded/dp/B017TFR664/ref=sxin	Only used one foot of wire. The pack is sold as 10 feet.	$1.00	$1.00
	Small Magents	6x3 mm Magnets	28	1		Hardware	No	Purchased	Amazon	VSKIZ	https://www.amazon.com/VSKIZ-Refrigerator-Multi-use-Neodymium-	Pack of 60 magnets. Only 28 were used.	$0.13	$3.73
			95		TOTAL PARTS								TOTAL PROJECT COST	$668.50